

Laboratory 6

Investigating signal averaging and the spatial-temporal properties of acoustic waves.

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MAE 170

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Abstract

The detection of sound waves can become increasingly complex in environments of high external noise. In acoustic fields like sonar imagery and signal processing, studying sound wave propagation can help address this concern. This lab aims to examine how we can help improve signal detection for noisy signals through averaging multiple measurements and investigating the way a wave behaves spatially and temporally through a medium. To study the effect of signal averaging in noise reduction, a fixed speaker emitted a single “click,” which was recorded by a fixed microphone. This process was repeated 64 times to obtain a representative average of the true signal. To assess the spatial-temporal propagation of waves, the same microphone was mounted on a moving arm that measured the signal at multiple different spatial positions. Experimental wave properties were calculated by finding the time delay between the reference and measured signals using single-peak detection and compared to theoretical values. The calculated speed of sound was found to fall within 5% of the theoretical value, demonstrating that averaging a noisy signal can refine signal clarity and, in turn, improve experimentally determined wave properties.

Introduction

Characterizing a sound wave’s properties in different media and understanding how to improve signal clarity in noise-heavy environments are essential parts of fields like fracture characterization, geosteering, and other subsurface imaging and material analysis applications [1]. The propagation of sound waves is caused by pressure fluctuations moving through a physical medium, where particles in a medium oscillate about their equilibrium positions [5]. Tracking these fluctuations across a medium as waves allows for sound to be analyzed. However, the readings of acoustic signals are easily affected by background noise and instrument fluctuations, which often hinder waveform clarity and reduce reading accuracy. [3]

A common way to reduce the detrimental effect of noise on signal clarity and accuracy is to average a signal over many measurements. Assuming the measurements are in near-identical conditions, the random noise tends to destructively interfere with itself while the reference signal becomes more apparent. Statistically, the uncertainty due to noise reduces by the square root of the total number of measurements taken: $\sqrt{N}$ [2],[4].

However, the actual extent to which signal averaging and spatial-temporal sampling improve acoustic signal clarity in noisy environments has not yet been properly quantified.

This experiment sought to characterize and quantify how effective these techniques are by calculating the speed of sound using a speaker and microphone system.

In the first part of the experiment, a speaker was placed at a fixed distance from a microphone and collected 64 different sound measurements. Averaging and comparing these measurements against the known reference signal, this part of the experiment was useful in testing the effectiveness of using signal averaging to reduce noise.

The second part of the experiment explored the spatial-temporal properties of the emitted sound waves. Here, the microphone was mounted to an arm that could move to discrete positions on the stage. The time delays between signal measurements recorded at various stage locations and the emitted reference signal were used to determine parameters of the emitted sound waves (speed of sound c , wavelength λ , and spatial resolution). The study of wave propagation showed that these properties are highly dependent on the geometric and material properties of the medium through which they are travelling [1],[5]. The combination of signal averaging and spatiotemporal measurements can work very well to produce more accurate measurements of sound waves when traveling through a medium.

Methods

Acoustic signals were generated as discrete pulses using an Arduino script-driven speaker fixed onto a 15 cm x 30 cm spatial stage, producing discrete impulses of 5kHz. The microphone was positioned at the origin of the coordinate system on the spatial stage, precisely located 30 cm right and 15 cm above the fixed speaker. All waveforms were recorded and visualized on a dual-channel oscilloscope at a sampling rate of, vertical scale of 50mV/div on the reference channel and 20mV/div on the microphone channel, and a rising edge trigger of 140mV with a normal sweep setting set to only capture discrete waveforms generated by the speaker at each position. Distance measurements were taken using a ruler placed on the stage with a resolution of 0.5 cm.

For the signal-averaging part of this experiment, a potential divider was connected between the Arduino “GND” pin and “9-pin” output channels to artificially reduce the output voltage of our Arduino and of each signal emitted. Stepping the output voltage made the measurement of the emitted signal more sensitive to noise, allowing the effects of signal averaging to be observed clearly. 64 discrete signals were emitted by the speaker and recorded by the microphone at a fixed position. These signals were then recorded on MATLAB and averaged to form an estimate of the reference signal. For the spatial-temporal investigation part of our experiment, the microphone, which was previously stationary, was moved in 1 cm increments across a grid using stepper motors, and a single signal was

measured at each position. Time delays between the emitted and measured signal were found using a peak-detection method through the use of MATLAB's tooltips.

Results

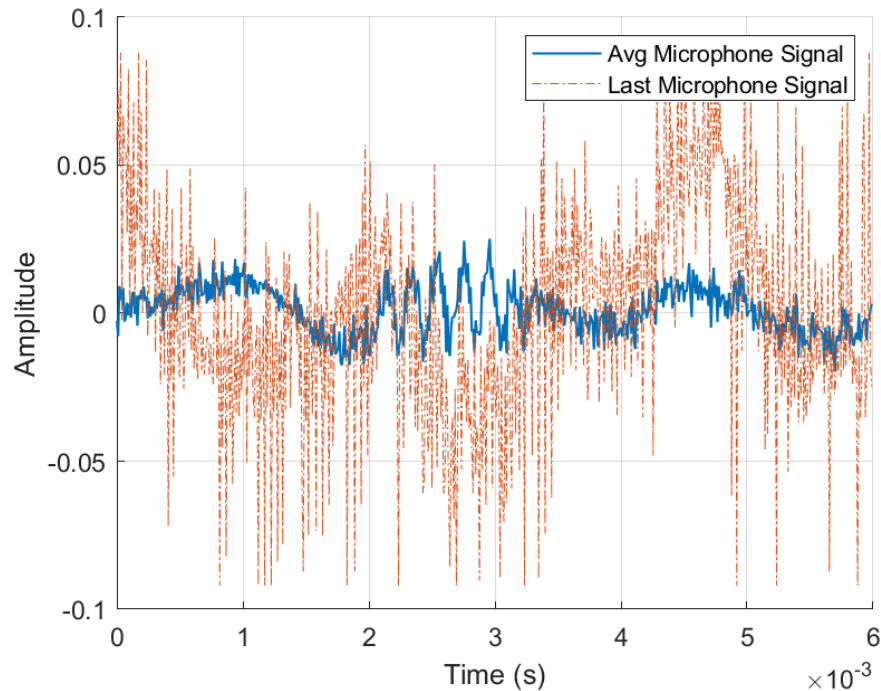


Figure 1. Amplitude of sound waves plotted across time. The blue solid line is the average of 64 signal measurements made by the microphone. The red dashed line is the last measurement made by the microphone.

In all work below, arrival times of a signal were determined by the peak of the first measured waveform to ensure consistency across all calculations.

Looking at the figure above, the last signal measured by the microphone seems very noisy, with wild oscillations from 0.1 to -0.1 in amplitude, and lacks clear peaks, making it difficult to interpret. The averaged signal, on the other hand, has much smaller oscillations, extending only up to around 0.025 in amplitude. Five clear peaks can also be identified between 2-3 milliseconds.

To investigate how averaging signals can be used to estimate original reference signals more in depth, **Figure 2** below shows how the averaged measured signal compares to the averaged reference signal.

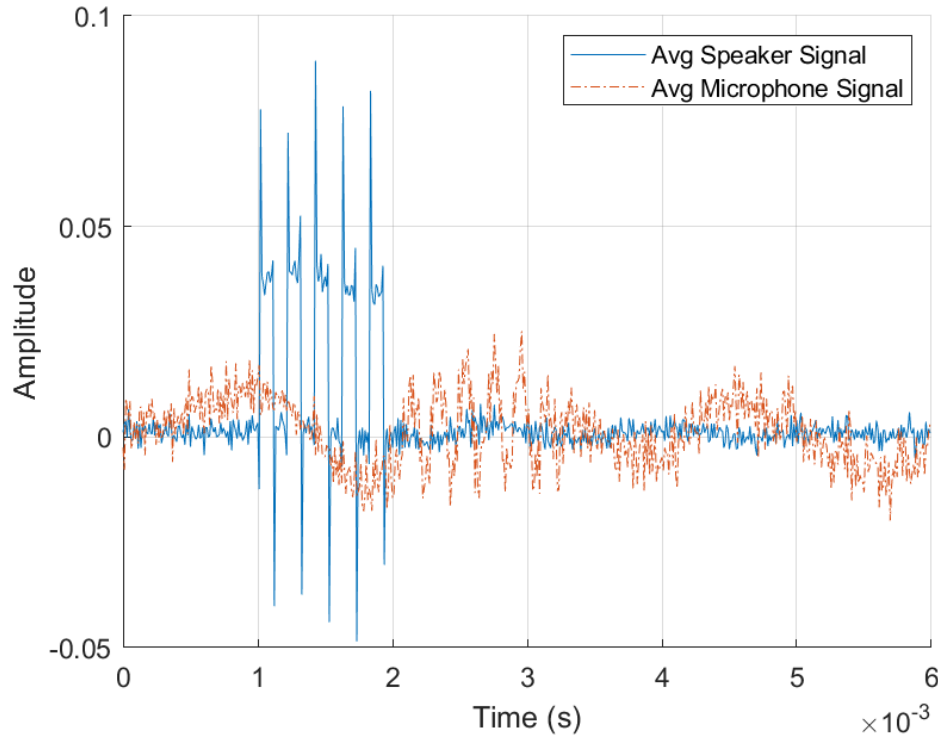


Figure 2. Amplitude of sound waves plotted across time. The blue solid line is the average of 64 signals emitted from the speaker. The red dashed line is the average of 64 signal measurements made by the microphone.

Looking at the figure above, the averaged reference signal from the speaker has five distinct peaks between 1-2 milliseconds, while the measured average microphone signal has five identifiable peaks between 2-3 milliseconds. Using the time delay between when the reference signal was first sent and when the microphone first received it, as well as the known location of the microphone relative to our speaker, an experimental value for the speed of sound in air can be determined. For the sake of calculation, the microphone position was designated as the origin.

Relative to our speaker, our microphone was located at these coordinates:

X-Distance (cm)	30 ± 0.5
Y-Distance (cm)	15 ± 0.5

Table 1: Microphone position relative to speaker in cm.

These measurements were obtained using a meter ruler, ensuring to measure from the center of the microphone to the center of the speaker. Even though the measurements were made with great detail and precision to avoid measurement errors, there was still an uncertainty of 0.5 cm. This came from sources of error including resolution, alignment, and human visual error.

The Pythagorean theorem was used to find the distance travelled by the sound wave between two points (from the speaker to the microphone). Defining our microphone to be at the origin, X_1 and Y_1 are both zero.

$$Distance = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2} \text{ (Equation 1)}$$

$$Distance = \sqrt{30^2 + 15^2}$$

$$Distance = \sqrt{1125} = 33.54 \text{ cm}$$

From the Pythagorean theorem, the straight line distance between the microphone and the speaker was calculated to be 33.54 cm or 0.3354 m.

The uncertainty in each ruler measurement was found to be half the smallest scale division, which for this ruler was 0.5cm. Equation 2 below was then used to find the uncertainty in the distance calculation:

$$Uncertainty = \sqrt{\left(\frac{\delta c}{\delta a} \Delta a\right)^2 + \left(\frac{\delta c}{\delta b} \Delta b\right)^2} \text{ (Equation 2)}$$

$$Uncertainty = \sqrt{\left(\frac{a}{\sqrt{a^2 + b^2}} \Delta a\right)^2 + \left(\frac{b}{\sqrt{a^2 + b^2}} \Delta b\right)^2}$$

$$Uncertainty = \sqrt{\left(\frac{30}{33.54} * 0.5\right)^2 + \left(\frac{15}{33.54} * 0.5\right)^2}$$

$$\text{Uncertainty} = 0.50 \text{ cm}$$

Using the straight line distance and uncertainty, the distance between the speaker and the microphone was calculated to be $33.54 \pm 0.5 \text{ cm}$.

Continuing this analysis, we found the time delay between when the output reference signal was sent out and when that signal was first received by the microphone. To do this, we used MATLAB's datatips tool in **Figure 2** to find the time at which the first peaks of the reference signal and measured signal occur. Seeing that the averaged measured signal was recorded at time 2.153ms and the average reference signal was first emitted at time 1.020, we can subtract the two measurements to find our delay:

$$t_{\text{delay}} = t_{\text{measured}} - t_{\text{reference}} \quad (\text{Equation 3})$$

$$t_{\text{delay}} = 2.062 \text{ ms} - 1.020 \text{ ms} = 1.042 \text{ ms}$$

The uncertainty of our measured signal due to our microphone is 0.1ms, or half of our signal's time period. We assigned it to be half of our time period because, with manual identification of peaks, uncertainty is half of the smallest scale resolution. We do not need an uncertainty for our reference signal as it is directly measured by the oscilloscope and assumed to be an exact reading.

Using our calculations of distance and time delay, we can find the speed of sound in air by the following:

$$a = \frac{\text{distance}}{\text{time}} \quad (\text{Equation 4})$$

$$a = \frac{0.3354 \text{ m}}{1.042 \times 10^{-3}} = 321.80 \text{ m/s}$$

We can then find the uncertainty of our speed of sound by:

$$\frac{\Delta a}{a} = \sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta t}{t}\right)^2} \quad (\text{Equation 5})$$

$$\frac{\Delta a}{a} = \sqrt{\left(\frac{0.5}{33.54}\right)^2 + \left(\frac{0.1}{1.042}\right)^2}$$

$$\frac{\Delta a}{a} = 0.0971$$

$$\Delta a = 0.0963 * 321.80 = 31.25 \text{ m/s}$$

Using this uncertainty as bounds on the theoretical speed of sound (340 m/s), we find the range below:

$$\text{Acceptable values: } 308.74 - 371.25 \text{ m/s}$$

$$\text{Percentage Error} = \frac{|Actual - Measured|}{Actual} * 100\% \text{ (Equation 6)}$$

$$\text{Percentage Error} = \frac{|340 - 321.8|}{340} * 100\% = 5.35\%$$

Our experimental value of 321.80 m/s falls well within our acceptable range of values and has a percent error of only 5.35% from the theoretical speed of sound, which we know to be 340 m/s.

The second part of the experiment focused on investigating the spatiotemporal response of the 5kHz signal emitted by the speaker. In the setup, the microphone was attached to an arm that was controlled by a stepper motor. Our microphone was located 30 cm to the right (x) and 15 cm above (y) of our fixed speaker, which we defined as the origin of our scene's coordinate frame. During the experiment, the arm moved in a 1 cm step-wise vertical snake pattern, alternating down and up while shifting left at the end of each cycle, ultimately configured to traverse 30 cm in x and 15 cm in y towards the speaker. At each discrete position, the oscilloscope captured both the measured and reference signal waveforms, which were then stored as matrices within MATLAB. Another distinction between part 1 and part 2 of the experiment is that this section did not use a potential divider because we were not interested in observing the effect of noise on our signal and were not conducting any signal noise averaging. Omitting the potential divider allowed us to get more accurate readings of our signal and reduce the intensity of noise interferences, as the magnitude of noise relative to our reference signal was much lower than in Part 1.

As the microphone moved around the scanning stage, we expected its actual movements and its commanded movements (from the MATLAB script) to be one-to-one, and thus end at the same position as the speaker. However, likely due to mechanical error from the motor,

there was a discrepancy both horizontally and vertically between our anticipated final positions and our expected final position.

	Expected Final Position	Actual Final Position
X position (cm)	0	9
Y position	0	4

Table 2. Tabulating our expected final position of our moving microphone vs our actual final position after the MATLAB script was completed.

Using the final positions measured, new motor step sizes were tabulated in the x and y directions to account for the error in position. The position ranges were also adjusted such that the motor traversed from 30 cm to 9 cm in the x-direction and 15 cm to 4 cm in the y-direction so that sound speed calculations would be accurate.

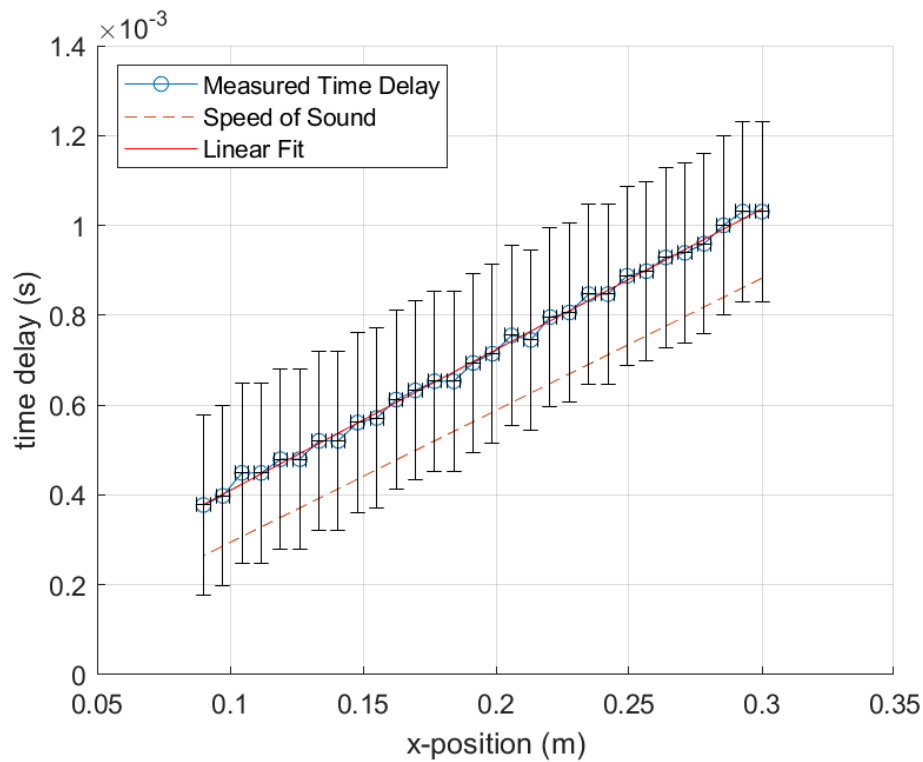


Figure 3. Measured delay (sec) against distance (m) for a 5kHz emitted signal being measured by a microphone at varying distances in the x-direction (9 cm - 30 cm). The y-position for these measurements is fixed at 10 cm. Blue circles represent measurement values, the orange dashed line represents how sound would theoretically travel, and the red solid line is a linear fit computed to the measurement values. Black bars in the x and y directions are uncertainty bars on the measurement values.

In the plot, uncertainty in the x-direction was set to be $\pm 2.5 \text{ mm}$ due to design aspects of the stepper motor. This motor used a belt with teeth that were spaced 2.5mm apart to rotate a shaft. This made the smallest possible step size 2.5mm, as the stepper motor simply slid to the nearest groove in the belt if it was within this bound.

Observing this figure, there seemed to be an offset between the linear fit and the speed of sound line of around 0.1 ms, but both have very similar slopes.

The polyfit function on MATLAB was used to find a line of best fit represented by the equation below.

$$y = mx + b; m = 0.0031, b = 9.6458 \times 10^{-5}$$

Looking at the units for the axes, the slope for this equation has units of time/distance. Since speed has units of distance/time, an estimate for the speed of sound can be found by taking the inverse of m.

$$a = \frac{1}{0.0031} = 322.58 \frac{\text{m}}{\text{s}}$$
$$\text{Percentage Error} = \frac{|340 - 322.58|}{340} * 100\% = 5.1\%$$

This value matches well with the calculation for the speed of sound in the previous part of the experiment, and only differs from the theoretical speed of sound by 5.1%.

To better understand how the sound was perceived by the speaker as it moved across the stage, the recorded audio was mapped with respect to the microphone's position as a function across time. The following 3 figures help visualize the sound waves emitted by the speaker.

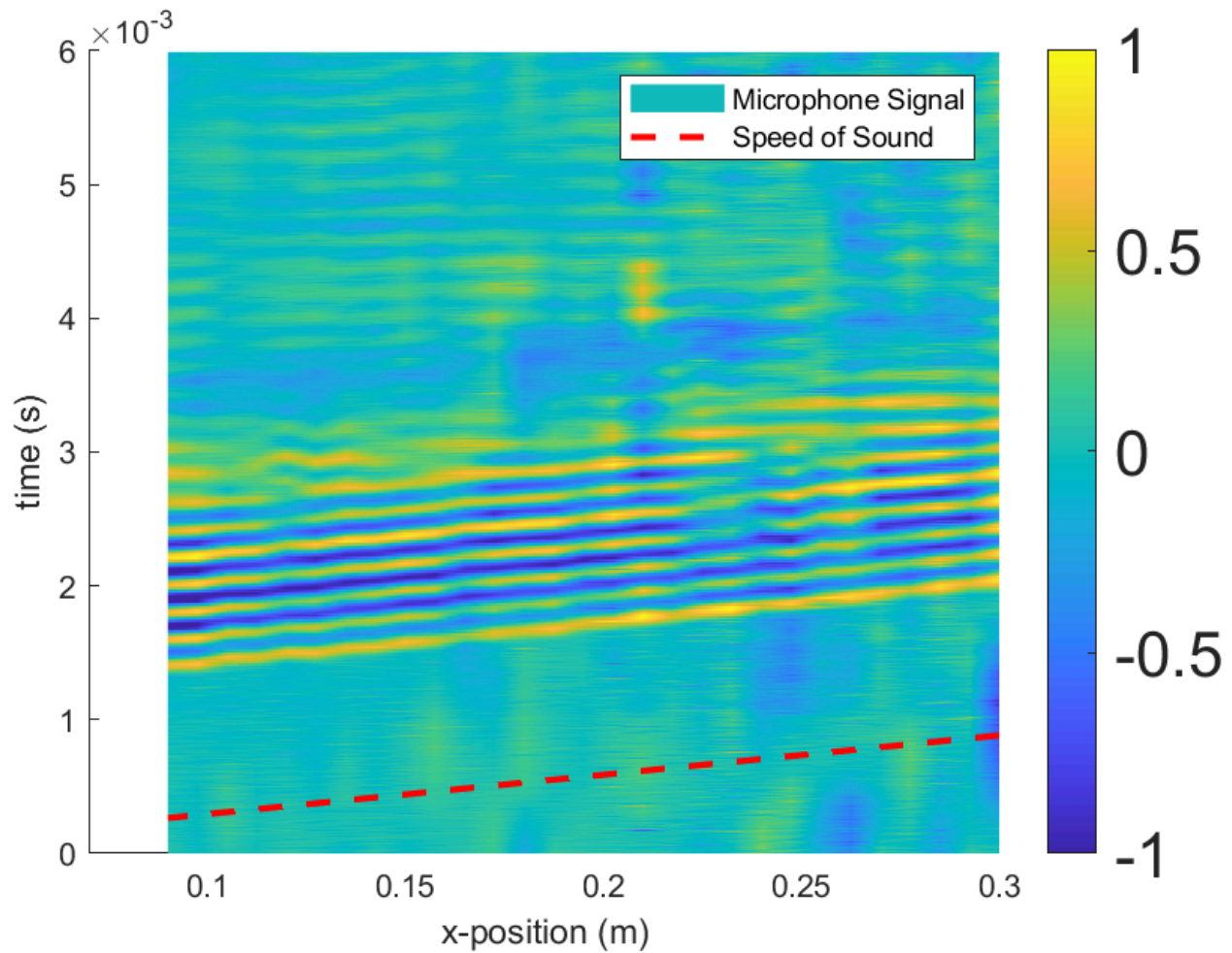


Figure 4. Amplitude of sound wave reaching the microphone across time at various x-positions. The y-position of the microphone during these readings is held constant at 10 cm. The amplitudes at a given location are normalized by the highest magnitude recorded across time such that the color scale on the right ranges from -1 to 1. The red dashed line corresponds to the speed of sound in air (340 m/s)

In this figure, the presence of a strong amplitude appears later in time as x-position of the microphone increases. Put together, the stronger amplitudes form sloped lines that are almost perfectly parallel to the speed of sound line.

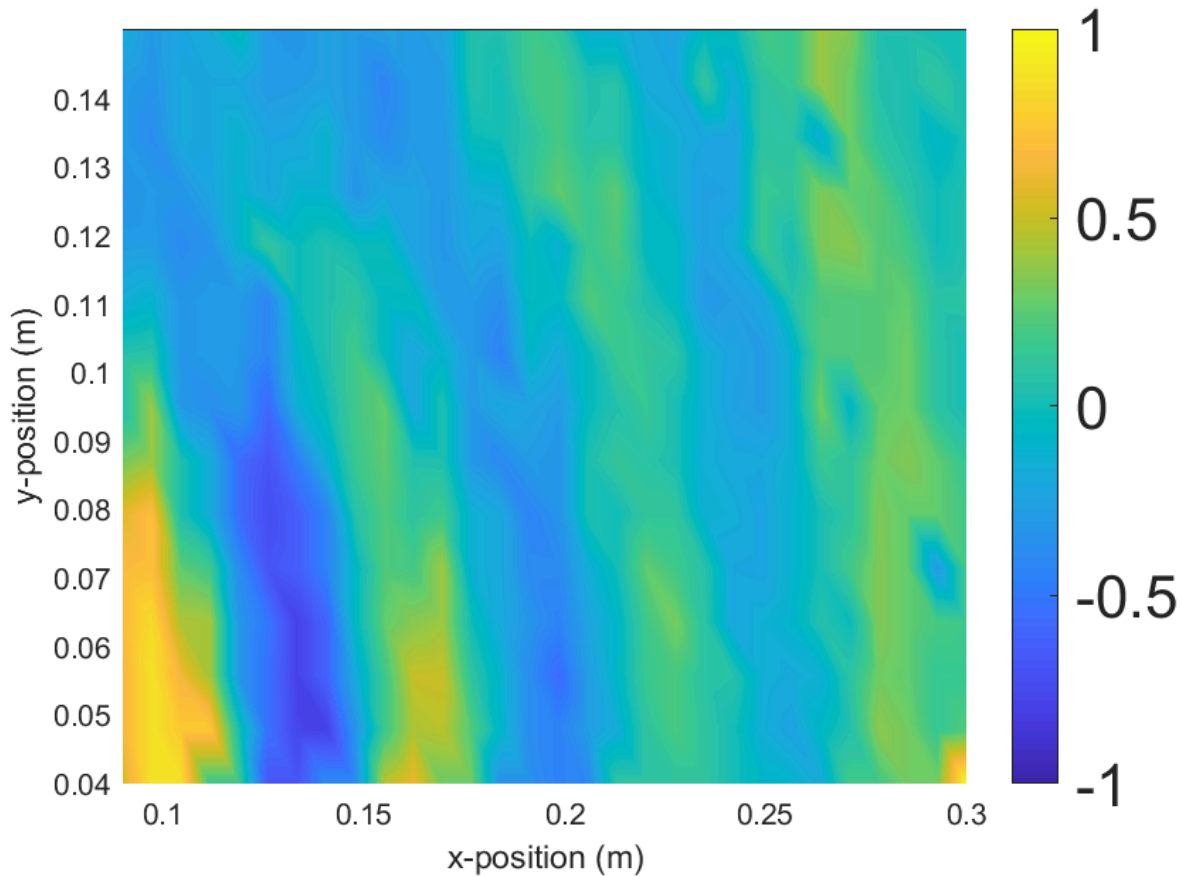


Figure 5. Snapshot of signals measured by the microphone at various locations on the stage at 2ms. The x-position varies from 9 cm to 30 cm, and the y-position varies from 4 cm to 15 cm. The bar on the right converts amplitude of the signal to a color that reflects signal strength on the grid.

Observing this image and paying close attention to areas of high magnitude, 4 curved vertical bands can be made out. In between these bands are regions of low amplitude signal.

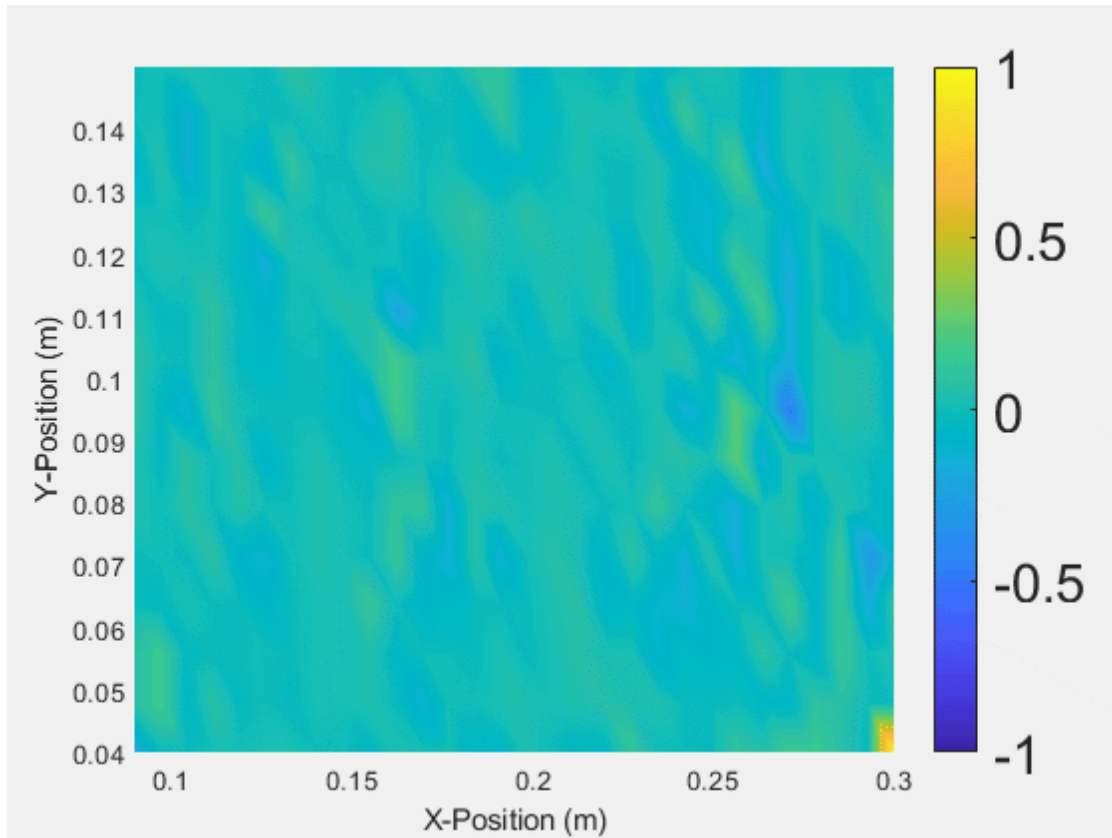


Figure 6. Animation of signals measured by the microphone at various stage locations from 0 to 6 ms. The x-position varies from 9 cm to 30 cm, and the y-position varies from 4 cm to 30 cm. The bar on the right converts amplitude of the signal to a color that reflects signal strength on the grid.

Building from the last figure, the curved vertical bands of higher magnitude seem to move across the image from left to right across time. The value of a position closer to the origin is mirrored by a position further from the origin at a later time.

In order to verify if the microphone movement increments ($10 \text{ mm} \pm 2.5 \text{ mm}$) were small enough to accurately measure the outputted speaker signal, the spatial resolution needed to be compared to the Nyquist sampling frequency.

The spatial resolution was found by dividing the motor step size by the acoustic wavelength:

$$\text{spatial resolution} = \frac{x}{\lambda} \text{ (Equation 8)}$$

The wavelength was calculated using the relation below from the speed of sound and the signal frequency:

$$\lambda = \frac{c}{f} \text{ (Equation 9)}$$

Accounting that the Arduino emits a signal at exactly 5kHz, and using the theoretically established value for the speed of sound to be 340 m/s, Equation 9 can be used to find the wavelength of the emitted signal:

$$\lambda = \frac{340 \text{ m/s}}{5000 \text{ 1/s}} = 0.068 \text{ m}$$
$$\lambda = 68 \text{ mm}$$

The spatial resolution can be found by using the wavelength above in Equation 8:

$$\text{spatial resolution} = 0.15$$

In order to sample a signal accurately, the Nyquist criterion states that at least two points must be sampled in one period. We can express this relationship by the equation:

$$\frac{x}{\lambda} \leq \frac{1}{2} \text{ (Equation 10)}$$

By Equation 11, the value of 0.15 is much lower than 0.5, ensuring that the spatial resolution in this experiment is low enough to measure the waveforms emitted by the speaker accurately using 1 cm discrete step sizes.

Recording the measurements taken in this experiment took a significant amount of time based on the MATLAB script commands and number of signal samples. Calculations on the run time for the script utilized in the second part of the lab can be found in **Appendix A** of our report.

Discussion

The first part of the experiment demonstrated how signal noise averaging can be used to clarify a noisy signal and better represent the original reference signal. Without the use of signal averaging, it could be shown in Figure 1 that there was seemingly a lot of random noise with rapidly oscillating amplitudes. With that, there was no clear waveform to indicate a reference signal being found. However, by averaging additional measurements, a much clearer signal was constructed. As seen in Figure 1, the 5 clear, discrete steps indicate that the signal emitted by our speaker was received by the microphone. Conceptually, it made sense that the noise was reduced by averaging many measurements because with each measurement of the signal, the noise within that signal was random. Therefore, as more measurements were taken, the random oscillations and amplitudes observed in the noisy signal interfered with each other destructively. The reference signal, on the other hand, remained the same and became more distinct amidst the noise-heavy environment.

Next, the time delay from when the reference signal was emitted by the speaker and detected by the microphone was calculated by analyzing peaks in the waveforms. The experimental value for the speed of sound in air was found to be inside of the range of acceptable values $340 \pm 28.64 \text{ m/s}$. This validates the use of signal averaging, because the speed of sound could not have been calculated from a single microphone signal. As stated before, the peaks would not have been easy to identify among the noise. Nonetheless, there was still some discrepancy between experimental and theoretical value, which was most likely due to the systematic errors in measurement. The errors in measuring likely came from a combination of the following: the smallest scale of our resolution was to the nearest centimeter, and there may also have been parallax error from when measuring the distance from above.

In the second part of this experiment, the spatial-temporal properties of sound waves were investigated. More specifically, how they radially propagate through a medium, which in this case is air. An experimental value for the speed of sound of our sound wave could be

determined, which was found to be 322.58 m/s. This fell very close to our calculated speed of sound in the previous part, further validating the results. This value gave a percentage error of 5.12% (Equation 6). The deviation between the theoretical value and the experimentally determined value could be attributed to various other sources of error.

Some possible sources of uncertainty were human measurement error, equipment delays, and environmental noise in the lab. However, a larger cause for error could have been the positional deviation of the microphone due to the stepper motors. It was known that the stepper motor, by the end of the script, was supposed to have moved our microphone 30 cm left and 15 cm down, placing it in the position (0,0). However, when the script ended, and the position of the microphone was measured, it was found that it had only moved to (9,4). This displays the large discrepancy in distance travelled in real life versus what the MATLAB script expects.

This type of uncertainty could be mitigated by changing the experimental setup in terms of the motors used to drive microphone movements. The stepper motors used an open-loop control system to operate the system, meaning that the controller always assumed that the motor had accurately executed each rotation. This meant that the controller had no feedback on the motor's position after the script had started, and thus would not know if the motor moved to the right position. This could lead to a compounding error, as the system could constantly be making small step errors in the microphone's position.

A better option for this experiment would be to use servo motors that operate with a closed-loop feedback system. This would allow the controller to receive the actual position of the motor after each movement and allow the controller to correct any discrepancies in movements. This would largely reduce sources of error in the experiment and make readings more accurate.

Observations from Figures 4 to 6 can help characterize the spatial-temporal behaviour of the 5 kHz sound wave. In Figure 4, the signal seemed to shift backwards in time as the x-position increased. The shifting of this band across space and time could be used to visualize the speed of sound as it tracked how the sound wave propagated. Seeing that this band had a slope that closely resembled a line representing the speed of sound, it is easy to see that the experiment effectively captured the propagation of the wave. Figure 5 can be used to visualize the lag of sound as x and y positions are increased, or as the microphone is placed further from the speaker. The skewing of the vertical bands to the left was indicative of locations at a further y-distance receiving the same signal data at a later time. The multiplicity of bands was indicative of locations at a further x-distance perceiving the same signal data later in time. Taking these observations and analyzing the animation in Figure 6,

it could be seen how sound waves propagated through the medium. It can also be readily observed how long it takes for sound to travel, noting the trend of a longer time for locations that were further away.

Conclusion

This lab successfully demonstrates how signal averaging can be used to significantly improve the detection and analysis of acoustic signals in noisy environments. The experiment also demonstrated how spatial-temporal measurements of acoustic signals can be used to characterize wave propagation through the air. By taking 64 measurements and averaging the signal of each measurement, environmental noise destructively interfered with itself and made the reference signal more distinct. Experimental speed of sound values of 321.8 m/s and 322.6m/s were within around 5% error of the theoretical value of 340 m/s.

Moreover, the spatial-temporal mapping of the acoustic signal helped display patterns of propagation and indicated that the acoustic wave was radially propagating as it travelled outwards, confirming the theory of a wave's radial propagation through a medium like air.

Overall, this experiment was able to highlight how signal averaging can be used to enhance signal clarity and improve reading accuracy for all acoustic system applications. This has applications in fields including sonar, medical ultrasound, and structural health monitoring, where signal averaging and spatial-temporal analysis are used to gather accurate readings in sub-optimal environments. Some future improvements that could be made to reduce uncertainty and increase the accuracy of the readings could be having a quieter surrounding environment, using servo closed-loop motors, and collecting more samples to improve the average approximations.

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Appendix

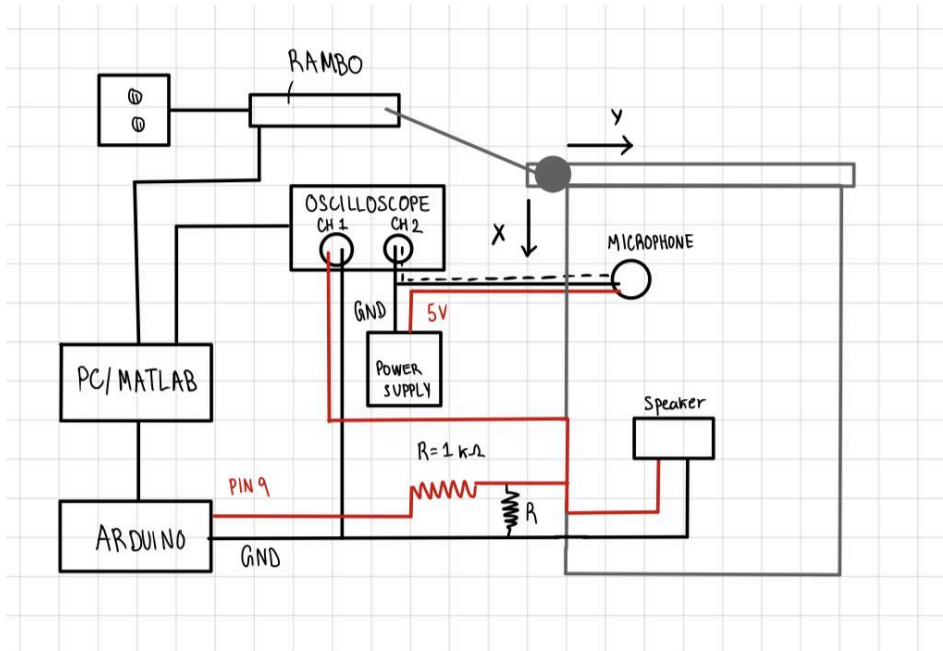


Figure A: Lab 6 Part 1 wiring diagram with voltage divider inserted between Arduino output and speaker.

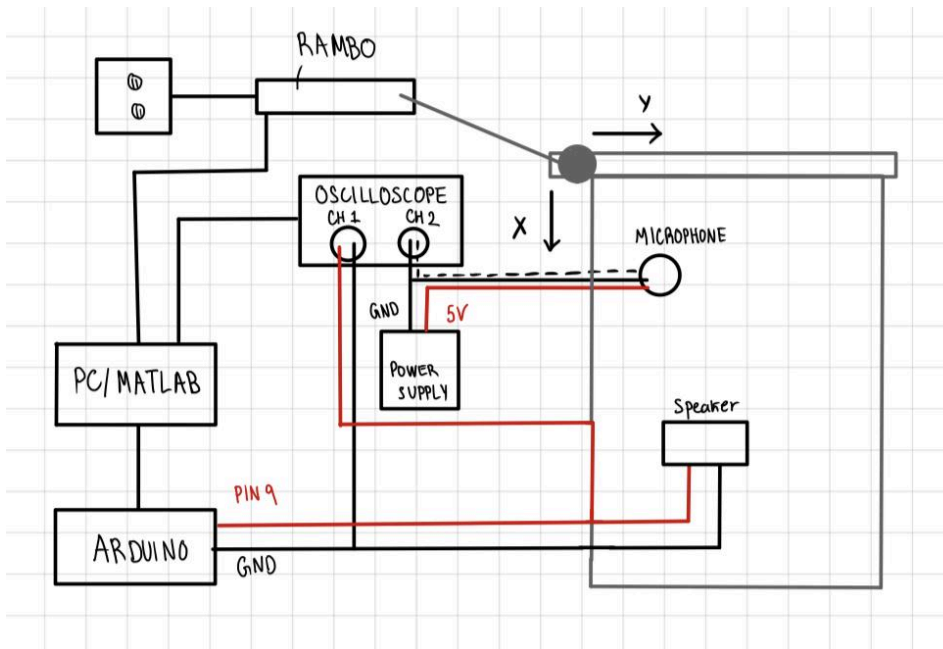


Figure B: Lab 6 Part 2 wiring diagram, without voltage divider.

Appendix A: Calculating the runtime of Part 2 script for averaging and individual sampling cases

In the second part of this experiment, 30 different x positions and 15 different y positions were measured along the spatial scene, making a total of 450 individual positions. In the script provided, there is a long initial pause (4 seconds) at point (1, 1) and subsequent points have a short pause (1 second) before the script continues onto the next measurement. It can be found that the total time a full cycle takes can be calculated by the equation:

$$total\ time = 4 + N_{short} \text{ (Equation 11)}$$

Since there were 450 individual positions, it is known that $N_{short} = 450$. The total time for one cycle came to:

$$\begin{aligned} total\ time &= 454\ s \\ total\ time &= \frac{454}{3600} = 0.12611\ hours \\ total\ time &= 0.13\ hours \end{aligned}$$

For a single sample at each position, the total run time is 0.13 hours. To find the total run time where 64 samples are taken at each position, N_{short} can be multiplied by the number of samples taken to get:

$$\begin{aligned} total\ time &= 4 + 64 \star 450 \\ total\ time &= 4 + 28800 = 28804\ s \end{aligned}$$

Converting this to hours:

$$\begin{aligned} total\ time &= \frac{28804}{3600} = 8.0011\ hours \\ total\ time &\sim 8.00\ hours \end{aligned}$$

This gives a total run time of 8 hours for taking 64 samples at each position along the 450 positions where the microphone is placed.

Appendix B: GIF Google Drive link - [■ sound_signal_greg_sainirnay.gif](#)

Greg's peer review

Reviewing Allen Vadukoot's Introduction

The introduction gives a good overview of the experiment's main topic, clearly explaining the spatial-temporal behavior of sound waves, the motivation behind the experiment, and the challenges associated with measuring acoustic signals. Moreover, it offers helpful background information on the phenomena being researched and provides a clear summary of the experimental setup, including the use of automated motor systems and signal-processing techniques. However, this section would benefit from including in-text references to support the claims being made throughout the introduction about spatial-temporal wave properties, measurement limitations, and signal noise averaging. While a clear knowledge gap statement is made for the spatial-temporal part of this experiment, there could also be a brief discussion on what the experiment aims to learn from signal noise averaging and how this can relate to the broader scientific community. Overall, this is a solid introduction with good structure, but it could be improved by using scientific sources to back up the claims made and expanding on the relevance of the entire experiment.

Reviewing Owen Hanenian's Results and Discussion

The results and discussion sections are able to effectively explain both Part 1 and Part 2 of the lab in good detail, including methodology, calculations, and explored phenomena. However, the report could use better structure and organization overall, particularly when it comes to presenting figures alongside their respective calculations. It would make this section easier to follow if the figures of Part 2 were shown after all of the Part 1 discussion was completed. Additionally, the figures are clearly labelled and well presented, but they all make use of both a title and a caption, which seems redundant and unnecessary. Another minor inconsistency was when the Green line of best fit mentioned in the discussion was not visible in the figure. The discussion of sources of error from stepper motors and improving on this error using servo-motors would be more effective by using interpretive phrasing like "a possible cause of error..." and "this suggests..." rather than using assertive

language. The parameter calculations are well written, but they should explain the motivation behind each calculation. They should explain where specific values come from (like the 1.1 factor) and assign more variables for clarity. Structurally, this section would improve by clearly separating Parts 1 and 2 into sections and replacing bullet points with coherent paragraphs to improve readability. Finally, the discussion should try to discuss things in context. For instance, when finding results, it would be helpful to tie the findings back to the abstract's problem statement to reinforce why the results are significant and to discuss whether they fall within the acceptable range.

Reviewing Bo Yan's Abstract

The abstract has a solid foundation and is written such that an engineer from another field could easily follow what is being researched; for instance, the example using Apple's headphone technology serves as a means to relate the experiment to the general public. However, the abstract could describe in more detail exactly what phenomena were studied in this experiment by giving more details as to what was actually being investigated. The abstract also lacks a concise summary of the main takeaways and results, which would help a reader understand the significance of this experiment and its conclusions. While the Apple example does its job in relating to the wider scientific community, the phrase "three-dimensional audio environments" lacks explanation, as the abstract does not explain exactly how this is relevant. The abstract would greatly benefit from making this experiment's main takeaways more explicit, such that the reader could infer them solely from reading the abstract. Finally, incorporating a brief passage on the methods used, like signal noise averaging and spatial-temporal mapping, would make this abstract more effective in communicating the overall methodology conducted in this experiment.

Sainirnay's peer review

Reviewing Shehreyar Asghar's Abstract

The abstract begins with a sentence introducing the topic, but this could be expanded on. It is not readily clear to the reader why or how exactly "characterizing acoustic wave propagation is fundamental" to engineering fields. Including another sentence or so to elaborate could improve the beginning of the abstract. Following the first sentence, a majority of the abstract section focuses on the approach followed in the lab. Though the methods section is long in comparison with the rest of the abstract, it does a good job of not going into too much detail, but it could be a bit more concise. The abstract concludes with a sentence commenting on the effectiveness of the lab, but this is not backed by a comprehensive summary of the most important results. It is left as an open-ended conclusion and not backed by any information from the experiment itself. This section could also be strengthened by including potential broader impacts and significant implications that extend to the real world. Overall, this abstract covers a majority of the points required on the rubric, but could use more detail in the background and implications portions. It also does a good job of maintaining past-tense language.

Reviewing Leila Juan's Conclusion

The conclusion begins by summarizing significant findings from the experiment performed and key takeaways from those results. The first paragraph includes numbers and percentages that provide a strong backing for the conclusions being drawn. The second paragraph could be strengthened by following a similar format so that the conclusions drawn about the Nyquist sampling criterion are self-evident. There are some grammatical errors in some of the sentences read, so fixing them will allow for better readability. Past summarizing results, the conclusion could be improved by expanding on future implications hinted at in earlier parts of the report. Discussing the significance of the results found would provide a meaningful conclusion that elaborates on potential future steps or experiments that can be taken to expand on this lab.

Reviewing Isabella Ly's Introduction

The first paragraph does a good job of introducing the topic and the physical phenomena being explored. Listing applications of where detection systems are important and citing relevant sources, it sets up the rest of the section and does a good job of following the rubric. However, the transition between the first and second paragraphs seems a bit abrupt. The flow could be improved by rearranging the first two sentences or so, leading with signal averaging and connecting that to the composition of a signal. The second and third paragraphs do a good job of setting up the purpose of the lab, but the knowledge gap statement required in the rubric is found at the very end. Structurally, this report could be improved by starting with the knowledge gap statement and following with the second and third paragraphs. This way, the summary of the work done feels responsive to the knowledge gap statement. Overall, this abstract does a good job of covering points on the rubric, but could be improved by rearranging certain parts for better readability and flow.